20261021 Quiz

for (MATH200) Differential Equations

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Problem 7.8.2. Find the general solution of the system

$$\mathbf{x}' = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix} \mathbf{x}.$$

Solution. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

The characteristic equation is

$$\det(\lambda \mathbf{I}_3 - \mathbf{A}) = (\lambda + 1)(\lambda - 2)^2 = 0.$$

For $\lambda_1 = -1$, solving $(\mathbf{A} + \mathbf{I}_3)\mathbf{v}_1 = \mathbf{0}$ gives

$$\mathbf{v}_1 = \begin{pmatrix} -3 \\ 4 \\ 2 \end{pmatrix}.$$

The repeated eigenvalue $\lambda_2 = 2$ has only one linearly independent eigenvector, since

$$\ker(\mathbf{A} - 2\mathbf{I}_3) = \mathrm{Span}\left(\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right).$$

Take $\mathbf{v}_2 = (0, 1, -1)^{\mathsf{T}}$. To obtain a third solution, find a generalized eigenvector $\mathbf{w}$ satisfying

$$(\mathbf{A} - 2\mathbf{I}_3)\mathbf{w} = \mathbf{v}_2.$$

For $\mathbf{w} = (w_1, w_2, w_3)^{\mathsf{T}}$, this equation becomes

$$\begin{pmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

One choice is $\mathbf{w} = (1, 0, 1)^{\mathsf{T}}$. The corresponding generalized-eigenvector solution is

$$e^{2t}(t\mathbf{v}_2 + \mathbf{w}),$$

and it is linearly independent of $e^{-t}\mathbf{v}_1$ and $e^{2t}\mathbf{v}_2$. Hence the general solution is

$$\boxed{\begin{aligned}\mathbf{x}(t) = c_1 e^{-t}\begin{pmatrix}-3\\4\\2\end{pmatrix} + c_2 e^{2t}\begin{pmatrix}0\\1\\-1\end{pmatrix}\\ + c_3 e^{2t}\left[t\begin{pmatrix}0\\1\\-1\end{pmatrix} + \begin{pmatrix}1\\0\\1\end{pmatrix}\right].\end{aligned}}$$

□

Problem 7.8.4. Find the general solution of the system

$$\mathbf{x}' = \begin{pmatrix}3 & 1\\-4 & -1\end{pmatrix}\mathbf{x}.$$

Solution. Let $\mathbf{A} = \begin{pmatrix}3 & 1\\-4 & -1\end{pmatrix}$. Its characteristic equation is

$$\det(\lambda\mathbf{I}_2 - \mathbf{A}) = (\lambda - 1)^2 = 0.$$

The eigenspace corresponding to $\lambda = 1$ is determined by

$$(\mathbf{A} - \mathbf{I}_2)\mathbf{v} = \mathbf{0}, \qquad \mathbf{A} - \mathbf{I}_2 = \begin{pmatrix}2 & 1\\-4 & -2\end{pmatrix}.$$

Thus it is one-dimensional, and we may take $\mathbf{v} = (-1, 2)^{\mathsf{T}}$. Since the repeated eigenvalue has only one independent eigenvector, choose a generalized eigenvector $\mathbf{w}$ such that

$$(\mathbf{A} - \mathbf{I}_2)\mathbf{w} = \mathbf{v}.$$

The equation $2w_1 + w_2 = -1$ is satisfied, for example, by $\mathbf{w} = (-1/2, 0)^{\mathsf{T}}$. Therefore $e^t\mathbf{v}$ and $e^t(t\mathbf{v} + \mathbf{w})$ form a fundamental set, and

$$\boxed{\begin{aligned}\mathbf{x}(t) = c_1 e^{t}\begin{pmatrix}-1\\2\end{pmatrix}\\ + c_2 e^{t}\left[t\begin{pmatrix}-1\\2\end{pmatrix} + \begin{pmatrix}-\frac{1}{2}\\0\end{pmatrix}\right].\end{aligned}}$$

□

Problem 7.8.6. Solve the initial value problem

$$\mathbf{x}' = \begin{pmatrix}3 & -1\\9 & -3\end{pmatrix}\mathbf{x}, \qquad \mathbf{x}(0) = \begin{pmatrix}2\\4\end{pmatrix}.$$

Solution. Let $\mathbf{A} = \begin{pmatrix}3 & -1\\9 & -3\end{pmatrix}$. The characteristic equation is

$$\det(\lambda\mathbf{I}_2 - \mathbf{A}) = \lambda^2 = 0.$$

The eigenvalue $\lambda = 0$ is repeated, and $\ker \mathbf{A}$ is one-dimensional. We may choose

$$\mathbf{v} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \qquad \mathbf{A}\mathbf{v} = \mathbf{0}.$$

A generalized eigenvector satisfies $\mathbf{A}\mathbf{w} = \mathbf{v}$. For example,

$$\mathbf{w} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \qquad \mathbf{A}\mathbf{w} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Since $e^{0t} = 1$, the general solution is

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + c_2 \left[t \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} \right].$$

Applying the initial condition at $t = 0$ gives

$$c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \qquad c_1 = 2, \quad c_2 = 2.$$

Hence the solution of the initial value problem is

$$\boxed{\mathbf{x}(t) = \begin{pmatrix} 2 + 2t \\ 4 + 6t \end{pmatrix}}.$$

□

Problem 7.9.3. Find the general solution of the system

$$\mathbf{x}' = \begin{pmatrix} 1 & 4 \\ 1 & -2 \end{pmatrix} \mathbf{x} + \begin{pmatrix} e^{-2t} \\ -2e^{t} \end{pmatrix}.$$

Solution. Write the system as $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$, where

$$\mathbf{A} = \begin{pmatrix} 1 & 4 \\ 1 & -2 \end{pmatrix}, \qquad \mathbf{g}(t) = \begin{pmatrix} e^{-2t} \\ -2e^{t} \end{pmatrix}.$$

The homogeneous system has characteristic equation

$$\det(\lambda \mathbf{I}_2 - \mathbf{A}) = (\lambda + 3)(\lambda - 2) = 0.$$

For $\lambda_1 = -3$ and $\lambda_2 = 2$, we may take the corresponding eigenvectors

$$\mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 4 \\ 1 \end{pmatrix}.$$

Thus the complementary solution is

$$\mathbf{x}_h(t) = c_1 e^{-3t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 4 \\ 1 \end{pmatrix}.$$

To find a particular solution by undetermined coefficients, note that neither -2 nor 1 is an eigenvalue of $\mathbf{A}$. Hence take

$$\mathbf{x}_p(t) = \mathbf{p} e^{-2t} + \mathbf{q} e^{t}$$

with constant vectors $\mathbf{p}$ and $\mathbf{q}$. Substitution into $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$ and comparison of the exponential terms give

$$(-2\mathbf{I}_2 - \mathbf{A})\mathbf{p} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad (\mathbf{I}_2 - \mathbf{A})\mathbf{q} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}.$$

Equivalently,

$$\begin{pmatrix} -3 & -4 \\ -1 & 0 \end{pmatrix} \mathbf{p} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & -4 \\ -1 & 3 \end{pmatrix} \mathbf{q} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}.$$

Solving these two systems gives

$$\mathbf{p} = \begin{pmatrix} 0 \\ -\frac{1}{4} \end{pmatrix}, \qquad \mathbf{q} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

Combining $\mathbf{x}_h$ and $\mathbf{x}_p$, we obtain

$$\boxed{\begin{aligned} \mathbf{x}(t) = c_1 e^{-3t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 4 \\ 1 \end{pmatrix} \\ + e^{-2t} \begin{pmatrix} 0 \\ -\frac{1}{4} \end{pmatrix} + e^{t} \begin{pmatrix} 2 \\ 0 \end{pmatrix}. \end{aligned}}$$

□

Problem 7.9.7. Find the general solution of the system

$$\mathbf{x}' = \begin{pmatrix} -\frac{5}{4} & \frac{3}{4} \\ \frac{3}{4} & -\frac{5}{4} \end{pmatrix} \mathbf{x} + \begin{pmatrix} 4t \\ e^t \end{pmatrix}.$$

Solution. Let

$$\mathbf{A} = \begin{pmatrix} -\frac{5}{4} & \frac{3}{4} \\ \frac{3}{4} & -\frac{5}{4} \end{pmatrix}, \qquad \mathbf{g}(t) = \begin{pmatrix} 4t \\ e^t \end{pmatrix}.$$

The characteristic equation of the corresponding homogeneous system is

$$\det(\lambda \mathbf{I}_2 - \mathbf{A}) = (\lambda + 2)\left(\lambda + \frac{1}{2}\right) = 0.$$

For $\lambda_1 = -2$ and $\lambda_2 = -1/2$, the corresponding eigenvectors can be chosen as

$$\mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Therefore the complementary solution is

$$\mathbf{x}_h(t) = c_1 e^{-2t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 e^{-t/2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

For a particular solution, the forcing term has a degree-one polynomial component and an e^t component. Since 0 and 1 are not eigenvalues of $\mathbf{A}$, use the trial

$$\mathbf{x}_p(t) = \mathbf{a}t + \mathbf{b} + \mathbf{d}e^t,$$

where $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{d}$ are constant vectors. Substituting and matching the coefficients of t, the constant term, and e^t gives

$$\mathbf{A}\mathbf{a} = -\begin{pmatrix} 4 \\ 0 \end{pmatrix}, \qquad \mathbf{A}\mathbf{b} = \mathbf{a}, \qquad (\mathbf{I}_2 - \mathbf{A})\mathbf{d} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Solving successively yields

$$\mathbf{a} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} -\frac{17}{2} \\ -\frac{15}{2} \end{pmatrix}, \qquad \mathbf{d} = \begin{pmatrix} \frac{1}{6} \\ \frac{1}{2} \end{pmatrix}.$$

Consequently the general solution is

$$\boxed{\begin{aligned} \mathbf{x}(t) = c_1 e^{-2t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 e^{-t/2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ + e^t \begin{pmatrix} \frac{1}{6} \\ \frac{1}{2} \end{pmatrix} + t \begin{pmatrix} 5 \\ 3 \end{pmatrix} + \begin{pmatrix} -\frac{17}{2} \\ -\frac{15}{2} \end{pmatrix}. \end{aligned}}$$

□